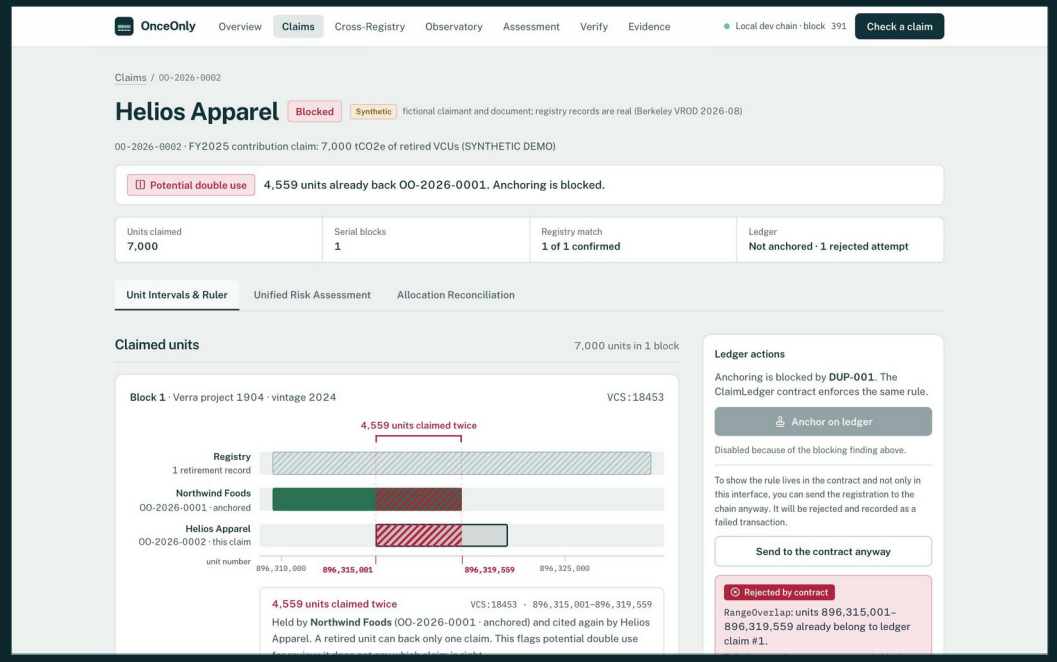
DATANOVA 2026 GLOBAL DATA-THON

OnceOnly

Every retired tonne, claimed once.

A carbon-credit claim integrity and investigation platform. It reconciles public climate claims with real registry retirements, explains every signal with its evidence and limits, and anchors claims on a contract that refuses to register the same unit twice.

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`github.com/Sidd927/OnceOnly-DataNova`



Real Verra retirement (Berkeley VROD 2026-08, batch 18453). Synthetic claimants. Local Hardhat development chain.

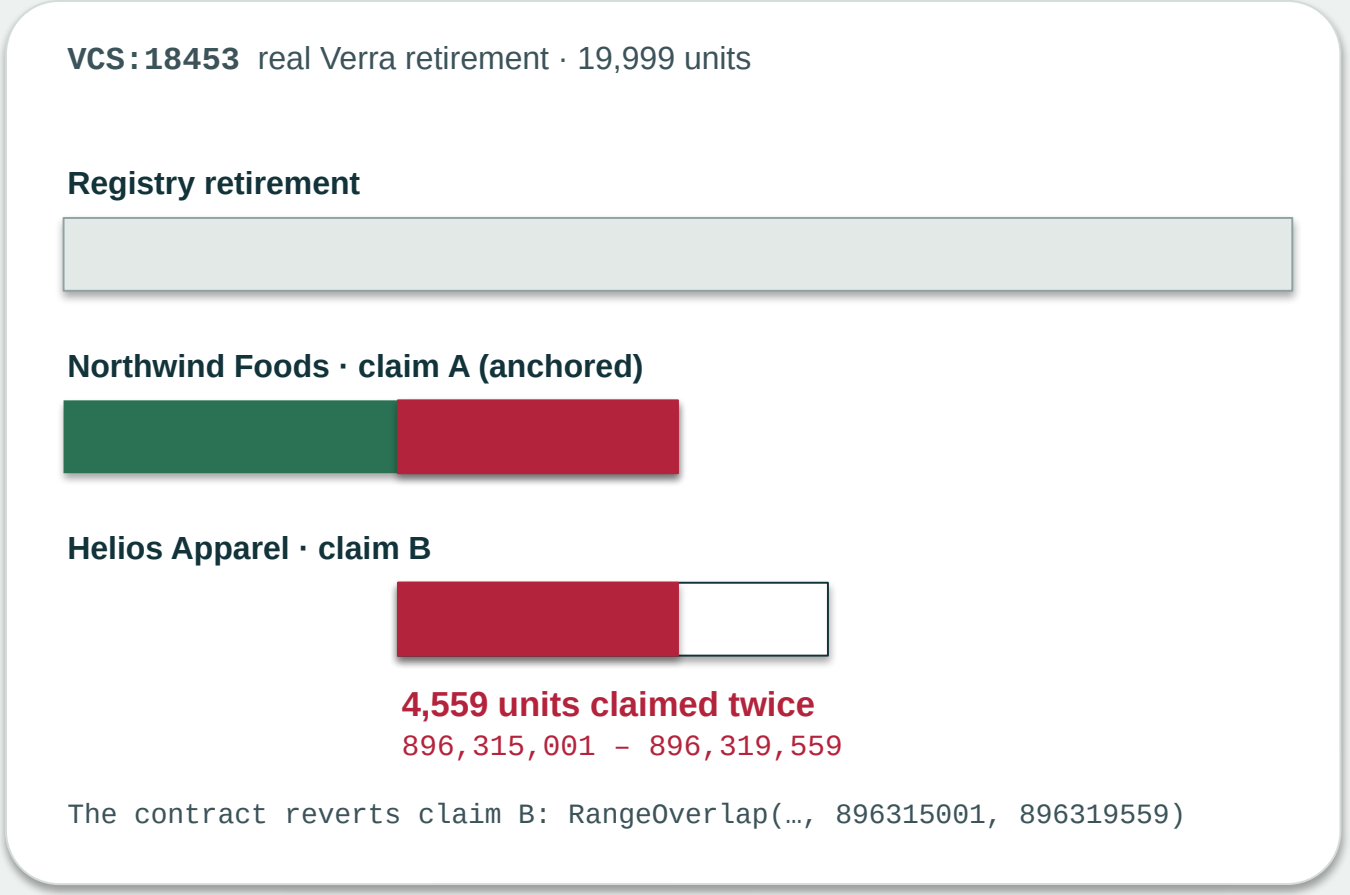
THE PROBLEM

A retirement happens once. Claims about it happen everywhere.

Registries stop a credit being retired twice. Nothing checks whether the same retired units then back two companies' public climate claims.

Brokers split one retirement across many buyers, and claims cite it in reports, labels and websites, each one valid when read alone.

Hashes and string matching miss it: overlapping claims usually cite different serial strings in different documents.



WHY SIMPLE CHECKS FAIL

The evidence is fragmented, and the obvious shortcuts break

0.333

recall of exact serial / document-hash matching

It misses every reformatted, partial, contained or superset overlap in the 270-pair evaluation.

159,511

spurious overlaps if the batch is ignored

Unit numbers restart in every issuance batch (11,631 Verra + 147,880 Gold Standard adjacent pairs). Ranges must be keyed by registry and batch.

62.7 %

of analysed retirements name no beneficiary

Undisclosed is a disclosure fact, not misconduct, but it limits who can be linked to which units.

2 roles

broker vs principal, parent vs subsidiary

"Retired on behalf of" and related companies look like conflicts to a naive interval check, and similar names are not legal identity.

Sources: eval/results/latest.json and registry_audit.json (full Berkeley VROD 2026-08: 493,518 rows); beneficiary share from the 37,335-retirement demo subset.

OUR SOLUTION

From a claim document to an explained, verifiable result

01
Read
Deterministic serial parser with provenance on every block; SHA-256 of each document

02
Reconcile
Serials become unit intervals in (registry, batch) namespaces, matched to real retirements

03
Allocate
Interval geometry + entity resolution: split, broker, related parties or conflict

04
Profile
Disclosure, concentration, fragmentation, timing and cross-registry context

05
Assess
16 deterministic rules, 6 scopes; evidence, counterevidence, missing data, next step

06
Anchor, verify
ClaimLedger refuses overlapping units; per-check public verification

WHO IT IS FOR

Sustainability auditors
check a client's claim against registry records and other claims

Registry / standards reviewers
see how retirements are split, disclosed and timed

Carbon-market analysts
triage projects, batches and entities with explainable signals

Highlighted steps (Allocate, Assess) are the DATANOVA additions to the original claim-ledger core.

ARCHITECTURE

Explainable analytics off-chain; one invariant on-chain

Data

· Berkeley VROD 2026-08 (real): 493,518 rows; demo subset 38,004 rows, 22 projects
· Non-destructive normalization: raw values kept
· Synthetic claim documents
· Curated benchmarks (124 + 36 cases)

FastAPI backend · Python 3.12

· Claim pipeline: serials → registry rules → findings → report hash
· 2A entity resolution: normalization, intermediaries, blocking, matching
· 2B allocation reconciliation: interval algebra, 8 allocation classes
· 2C concentration · fragmentation · temporal · cross-registry
· Stage 4 assessment: 16 rules, dependency groups
· Verification: on-chain vs off-chain checks reported separately
· SQLite (claims, read-only registry DB) · content-addressed evidence

ClaimLedger.sol

· Ranges keyed by keccak(registry:batch)
· O(1) hinted sorted list; RangeOverlap revert
· Local Hardhat chain 31337

React 19 UI

· Overview · Claims · Cross-Registry
· Observatory · Assessment
· Verify · Evidence
· Bespoke SVG charts, no chart library

Deployment: one Docker image (private chain, contract, seeded demo, API serving the UI), about 280 MB of memory; Render blueprint prepared.

TECHNICAL DIFFERENTIATION

Intervals, not hashes. Classes, not one red flag.

Allocation-aware reconciliation

Six geometric and eight allocation classes separate a legitimate split, a broker acting for a disclosed principal, related companies and an unresolved beneficiary from a conflict between unrelated claimants.

Namespace isolation

Every comparison is inside one (registry, batch). Identical numbers or project IDs across registries are an identifier collision, never the same asset.

Defensible statistics

HHI and entropy over disclosed volume only; pooled vs within-batch Gini reported separately; leave-one-out MAD bursts; absolute vs relative small allocations.

Explainability by construction

16 versioned deterministic rules; each finding carries thresholds, counterevidence, missing data and a next step; correlated signals grouped so they cannot double-count.

Verification you can redo

Document, commitment, ranges, network and deployed bytecode checked separately; database lookups never counted as on-chain proof.

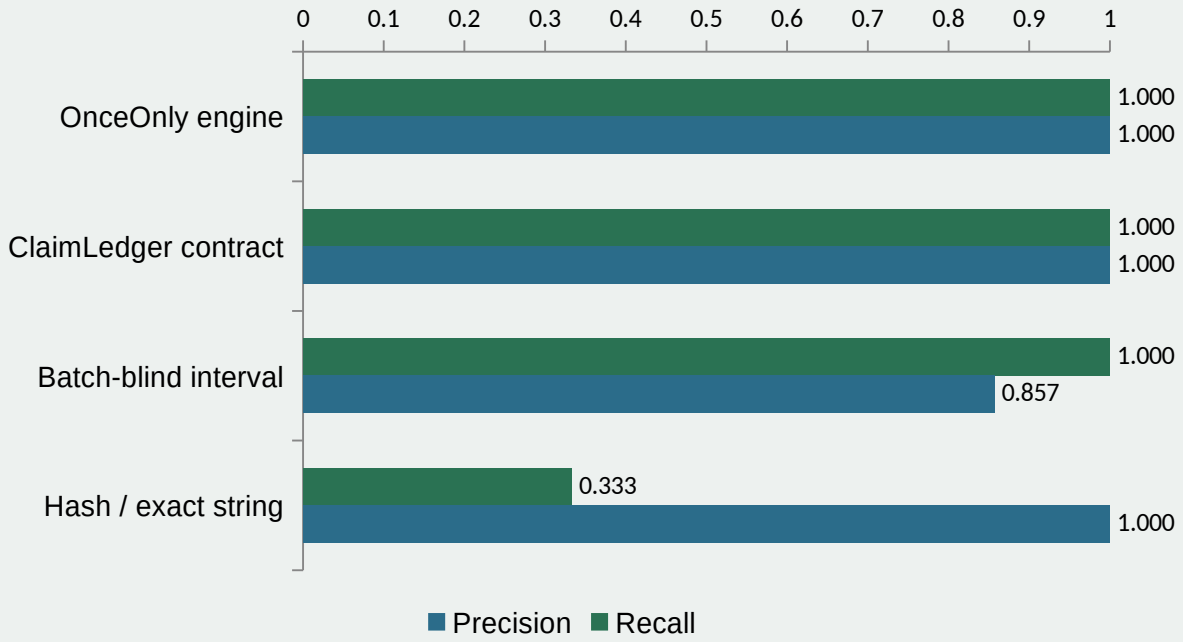
A contract that scales

Registration gas flat at 301,452 from 10 to 200 active ranges per batch (linear v0: 832,608); a bad hint reverts, it never skips the check.

EVIDENCE AND VALIDATION

Measured, and labelled for what each test measures

Double use on 270 synthetic claim pairs citing real Verra ranges



Ground truth uses the engine's interval definition, so 1.000 verifies the implementation; the finding is what the simpler methods miss. Not real-world accuracy.

Data and analysis checks

Registry serials → unit ranges (full VROD)	**493,518 / 493,518**
Overlaps inside one batch (Verra)	**0 of 307,416**
Entity resolution, 124 curated pairs	**P 1.000 · R 0.964**
Allocation class, 36 curated scenarios	**36 / 36**
Serial extraction, 60 synthetic snippets	**100 % in-sample**
Population metrics, independent SQL re-run	**exact match**

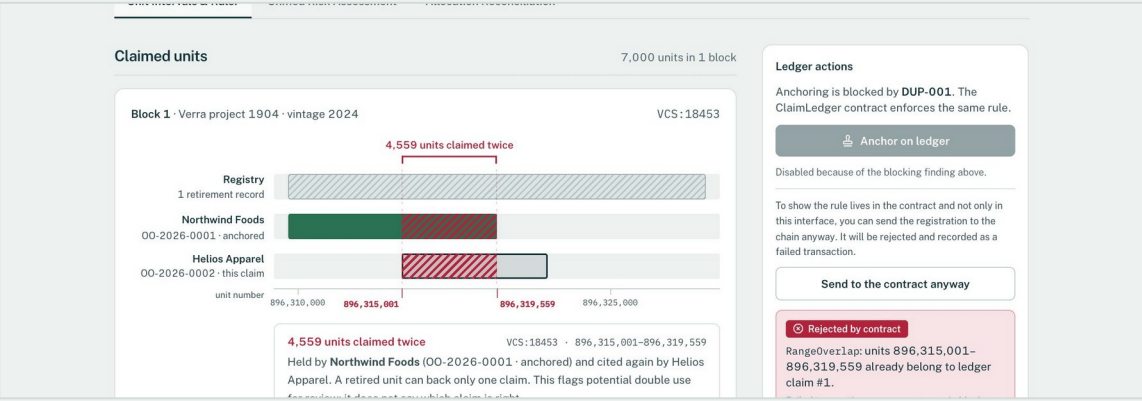
Software tests (this release)

274 backend · **24** contract · **45** browser · **13/13** end-to-end

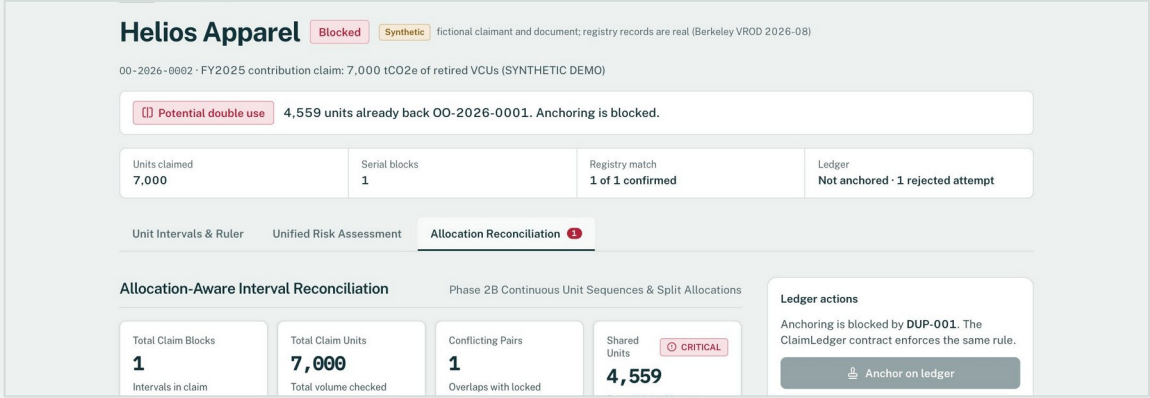
`Demo state fingerprint reproduced: c544f589…6483b`

PRODUCT DEMONSTRATION

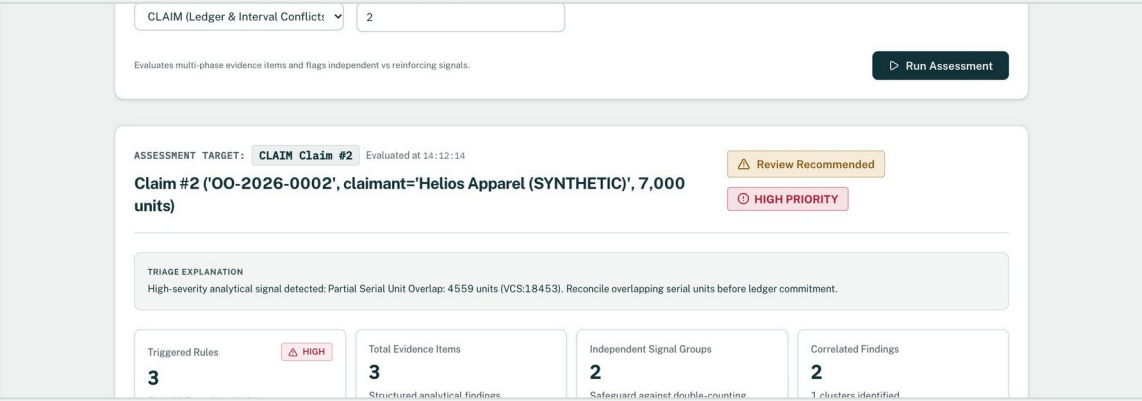
One real retirement, two synthetic claims, every step explained



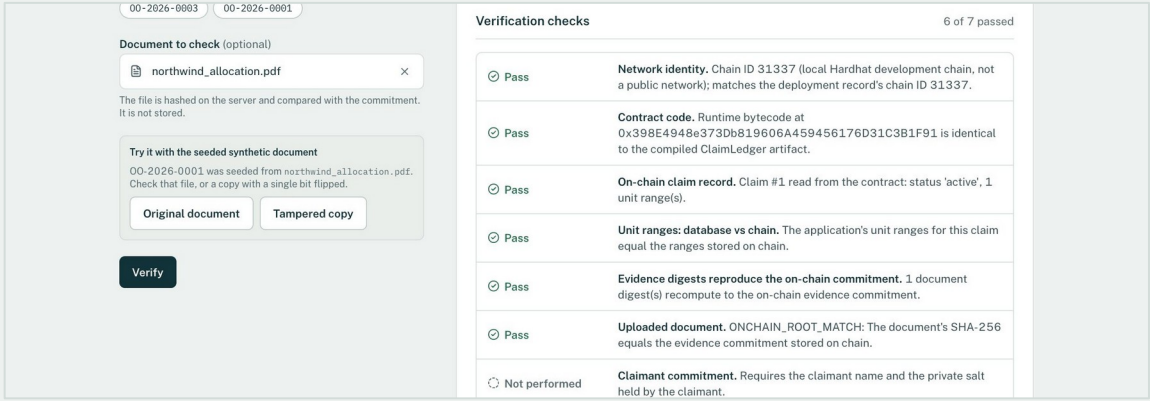
1 · The Ruler: 4,559 units already back Northwind's anchored claim; the contract rejects the second registration



2 · Allocation reconciliation: ALLOCATION_CONFLICT between different entities, with a next step



3 · Assessment: RULE-REC-002 with metrics, limitations and grouped concentration signals



4 · Verification: seven separate checks, including deployed bytecode and the document commitment

What it verifies, what it flags, what it cannot conclude

VERIFIES

· Serial strings parse into exact unit ranges and match registry retirement records (snapshot 2026-08-31)

· Two claims share specific units in the same registry batch

· A document is byte-identical to what was committed on chain

· The deployed contract code matches the compiled artifact

FLAGS FOR REVIEW

· Potential double use, classified by allocation type

· Concentration, fragmentation, timing anomalies

· Candidate cross-registry project links

· Missing beneficiary information (reported, never imputed)

CANNOT CONCLUDE

· Fraud, intent, or which claim is legitimate

· Legal identity from similar names or parent/subsidiary links

· Credit quality, additionality or permanence

· Anything about the whole market: analytics cover a 22-project subset

Trust boundaries: local Hardhat development chain (31337), not a public network · one trusted registrar key, no claimant signatures · synthetic claimants · static registry snapshot · AI extraction implemented but not measured · benchmarks synthetic or curated, not real-world accuracy.

IMPACT AND NEXT STEPS

From a working prototype to a usable integrity check

Practical value

Catches what document review and hashing miss: partial and reformatted overlaps of the same retired units.
Makes analysis reviewable: every signal shows its rule, threshold, counterevidence and next step, so an auditor can defend or reject it.
Reproducible: public data, open code, one-command container, deterministic demo state.
Neutral ordering: a contract invariant that the operator's own UI cannot bypass.

Credible path forward

1. Public testnet deployment with claimant-signed registrations (EIP-712)
2. Per-document Merkle commitments and proofs
3. Registry data beyond the 2026-08 snapshot; more registries
4. Measure AI extraction (accuracy, cost, latency) on real documents
5. Pilot with real published claims, with consent, to measure prevalence

Run it: `make docker-build && make docker-run → http://127.0.0.1:7860`
Code, data notes and methodology: `github.com/Sidd927/OnceOnly-DataNova` · Registry data: Berkeley Carbon Trading Project, VROD v2026-08 (CC BY 4.0)